WEEK NOTES · WEEK 4

AUTOMATE WITH AI

Voice agents, routines & workflows.

You were producing outputs. This week you build **infrastructure** — voice agents, Claude Routines, agentic loops and automated pipelines that reason, decide and act without you touching a keyboard.

THE IDENTITY SHIFT

AI Creative Director → AI Automation Architect → AI Systems Operator

THIS WEEK'S SHIFT

From making things with AI to building systems that run themselves.

Weekly objective

Build infrastructure — voice agents, Claude Routines, agentic loops and automated pipelines that reason, decide and act without you.

CLAUDE FEATURES APPLIED THIS WEEK

Routines — multi-step automated sequences.

Schedules — time-triggered automation.

MCPs — external tool access inside agents.

Claude API — Claude as the automation brain.

Projects & Skills — agent + voice-persona config; reusable agent logic and script templates.

Connectors — Notion, Slack, Gmail as data destinations.

TWO SESSIONS THIS WEEK

SESSION 1

AI Voice Technology + Voice Agents — from STT to live callable agents.

SESSION 2

Agents, Routines & Workflow Automation at scale — N8N, Make & Claude.

SESSION 1 – VOICE

How voice works end to end, and the platforms that make a callable agent.

SPEECH-TO-TEXT (STT)

How it works — acoustic modelling + language modelling + beam-search decoding turn speech into text.

Whisper (OpenAI) — multilingual, several model sizes; pick a size by accuracy vs speed.

Deepgram — real-time streaming STT, speaker diarisation, confidence scores.

AssemblyAI — STT with understanding built in: entities, sentiment, topics, safety.

TEXT-TO-SPEECH (TTS) & PROSODY

Synthesis — concatenative vs neural vs diffusion; neural sounds human.

Prosody — pitch, rate, emphasis and pauses, controlled with SSML markup.

VOICE CLONING

How — speaker embeddings map a voice into a latent space; zero-shot vs few-shot cloning.

ElevenLabs Voice Lab — clone from samples, design voices from acoustic parameters, multilingual.

PlayHT — ultra-realistic TTS, cloning, and a streaming API.

Consent — only clone a real person's voice with their written permission.

THE VOICE-AGENT STACK & CONVERSATION DESIGN

The stack — STT → LLM reasoning → TTS → telephony/channel → back to STT.

Conversation design — intents, entities, slots, dialogue states, flow diagrams.

Turn-taking — barge-in detection, silence thresholds, confirmations, graceful fallbacks.

Platforms — VAPI (numbers, webhooks, Claude, analytics), Bland AI (outbound), Retell AI (flows + persona).

SESSION 2 – AGENTS

What an agent really is, and how to automate one at scale.

WHAT AN AGENT IS

The loop — Perception → Reasoning → Action → Memory → repeat. Not a chatbot.

Four components — an LLM brain, tool access, a memory layer, and a planning mechanism.

Tool calling — Claude decides to call a tool, reads the response, and integrates the result.

Memory types — in-context, external (Notion/DB), semantic (vectors), episodic (past runs).

Failure modes — hallucinated tool calls, infinite loops, context overflow, instruction drift — design to prevent each.

CLAUDE ROUTINES, SCHEDULES & MCPS

Routines — multi-step prompt chains that run as one automated sequence (research → summarise → format → store → notify).

Schedules — recurring agents (daily brief, weekly digest) — set once, runs forever.

MCPs — give Claude real-time access to Notion, Slack, Gmail and Calendar inside a Routine.

Agent system prompt — goal, tools list, constraints, memory instructions, escalation logic.

WORKFLOW AUTOMATION AT SCALE

The model — triggers → actions → conditions → data transformation.

N8N — nodes, webhooks, HTTP requests, credentials, error handling and retries.

Claude in N8N — the HTTP Request node calls the Claude API; prompt for strict JSON; IF/Switch routing.

Make & Zapier — visual canvases, routers/aggregators, multi-step Zaps — pick per job.

Airtable — a structured data layer for tracking, filtering and dashboards.

Patterns — content pipeline, lead qualification, research digest, social scheduler, email router; voice → automation via VAPI.

THE THREE MODELS TO REMEMBER

1 · THE AGENT LOOP

Perception → Reasoning → Action → Memory ↻

An agent senses, thinks, acts, remembers — and repeats. That loop is what separates an agent from a chatbot.

2 · THE VOICE-AGENT STACK

STT → LLM reasoning → TTS → Channel → STT

Speech in, text out, Claude decides, speech back — over a phone line or app, then round again.

3 · THE 5-STEP ROUTINE

Research → Summarise → Format (JSON) → Store (Notion) → Notify (Slack)

The canonical Claude Routine — five steps, no code, running on a schedule.

YOUR TOOLKIT THIS WEEK

BRAIN

Claude — Projects, Routines, Schedules, MCPs, Connectors, and the API as your automation brain.

VOICE

STT — Whisper · Deepgram · AssemblyAI.

TTS & cloning — ElevenLabs · PlayHT.

Voice agents — VAPI · Bland AI · Retell AI.

AUTOMATION & DATA

Workflows — N8N · Make · Zapier.

Data — Airtable · Notion · Slack · Perplexity.

By the end of the week

You understand how agents, voice and automation actually work — from the agentic loop to the N8N node — and you have a live, running Automated AI System with a voice layer. The **AI Systems Operator** identity is unlocked.